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Investigating the Origin of UV Variability in Be/X-ray Binaries

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Be/X-ray Binaries: The Standard Theory

- Be star + Neutron Star Compact Object
- Moderately eccentric orbit
- Orbital Period: 10s - 100s of days

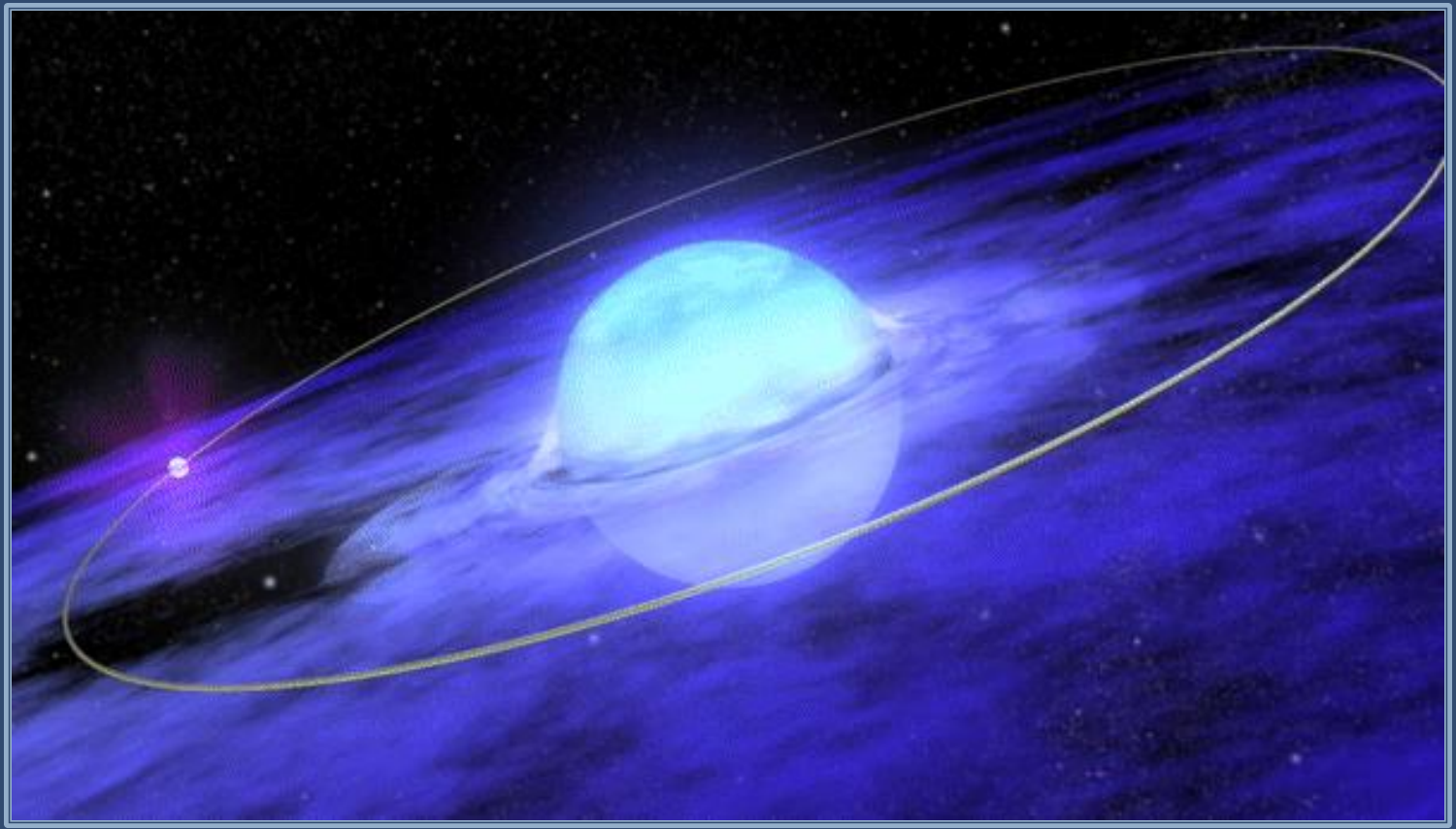


Image Credit: Walt Feimer, NASA/Goddard Space Flight Center

Be/X-ray Binaries: The Standard Theory

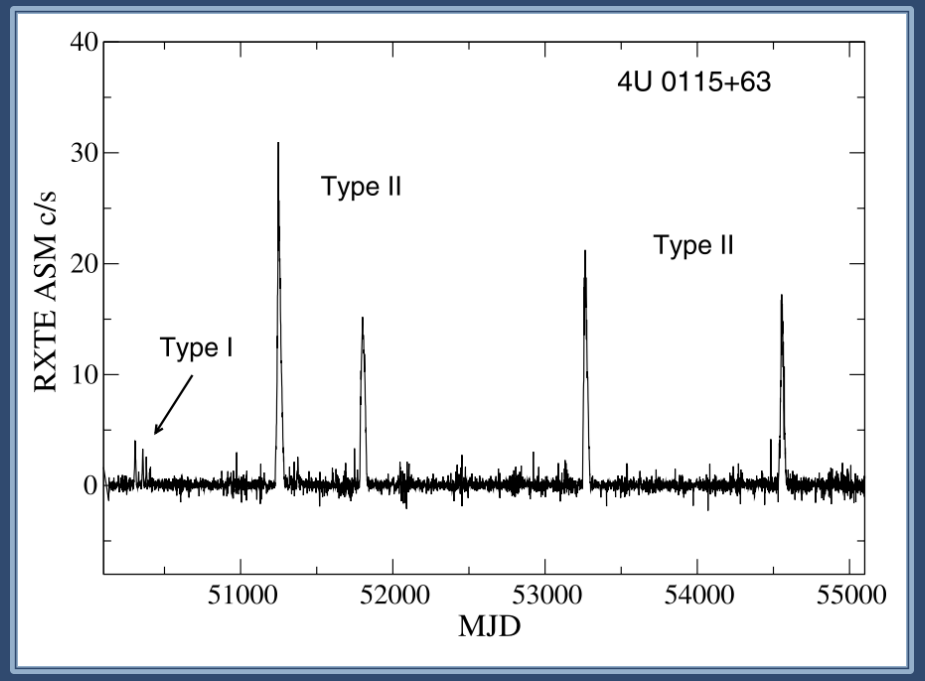


Image Credit: Reig 2011

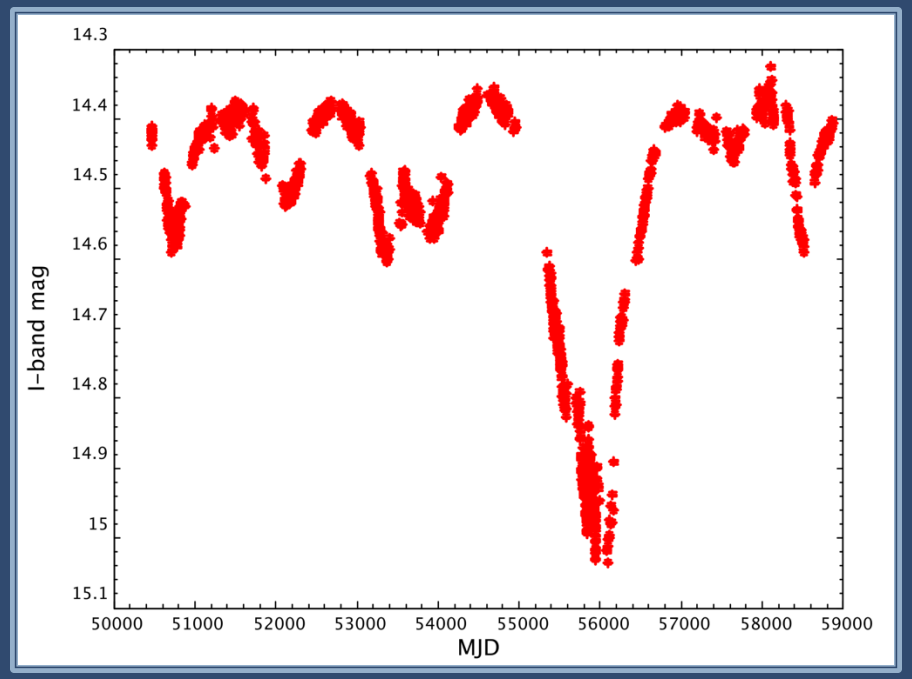


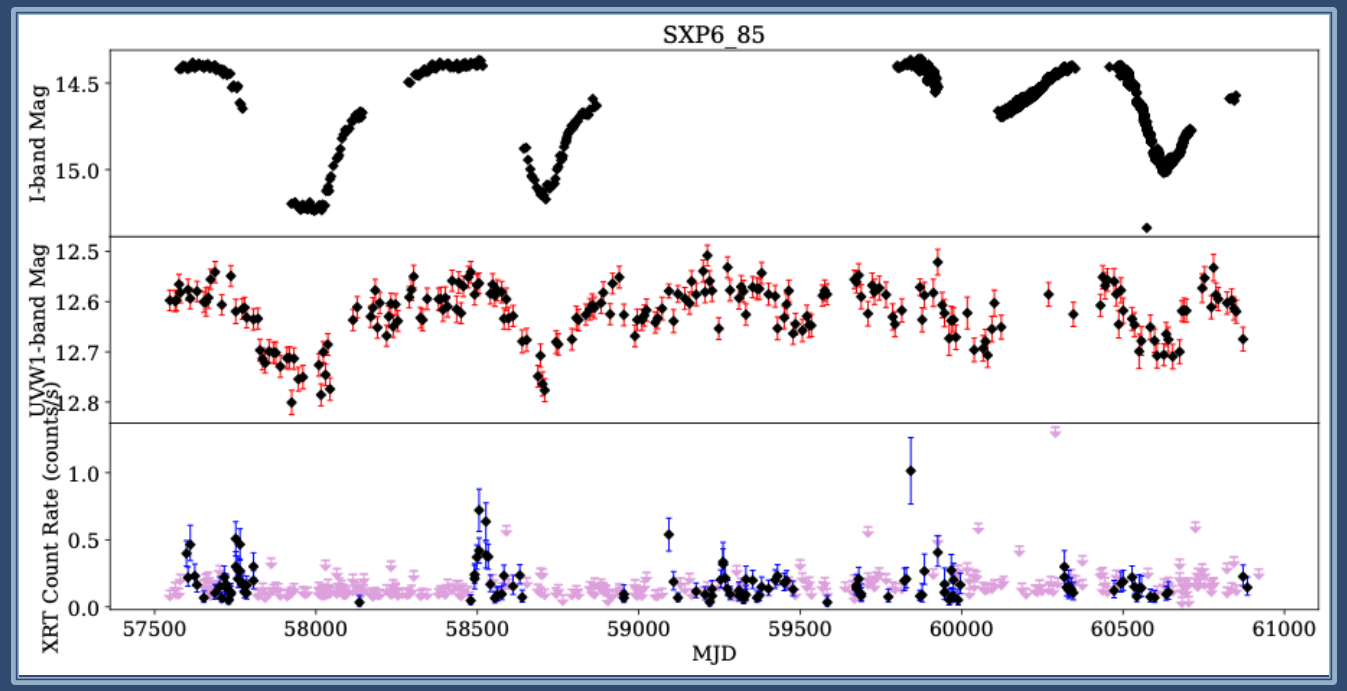
Image Credit: Monageng et al. 2022

- Transient X-ray emitters
- 2 Types of X-ray Outbursts

- Highly Variable in Optical/IR
- Variable on 3 different timescales: 0-2 Days, 2-200 days, 200+ days

Swift Reveals UV Variability

- Many BeXRBs are UV variable
- Variability is often coordinated with IR
  - Can be correlated or anit-correlated with X-ray
  - Sometimes observe lags between IR/UV
- Diverse behaviors observed from SMC population



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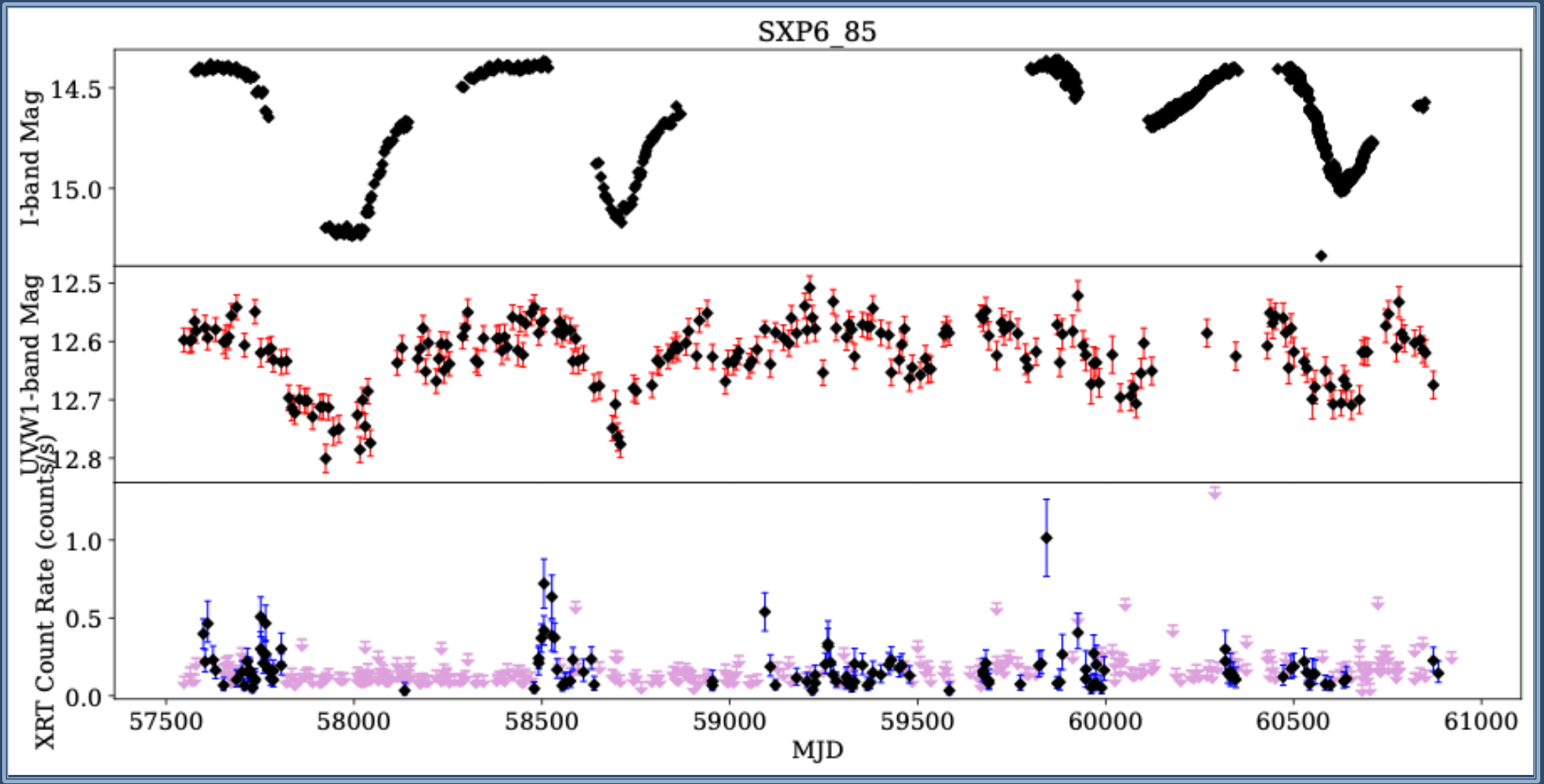

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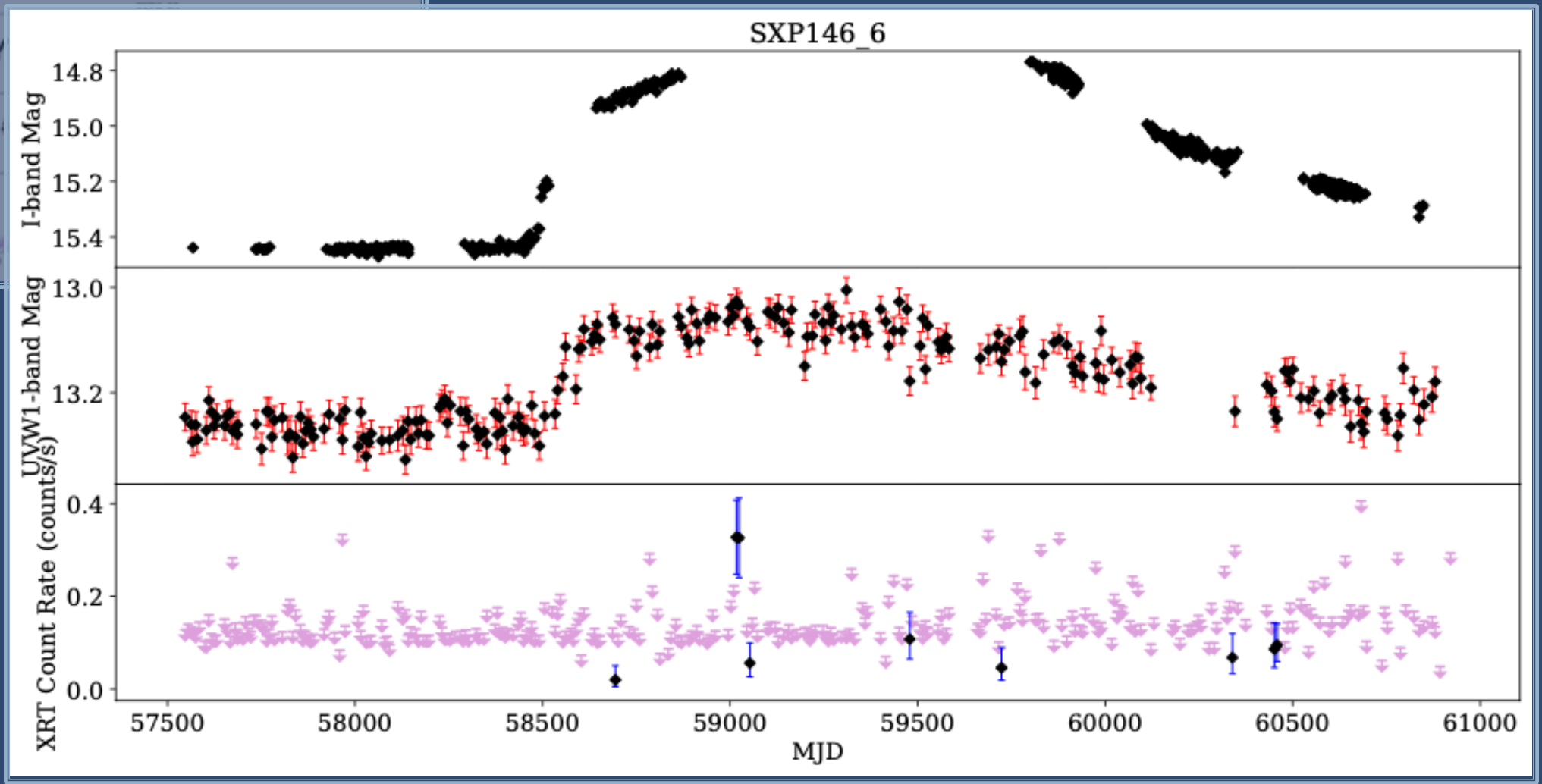

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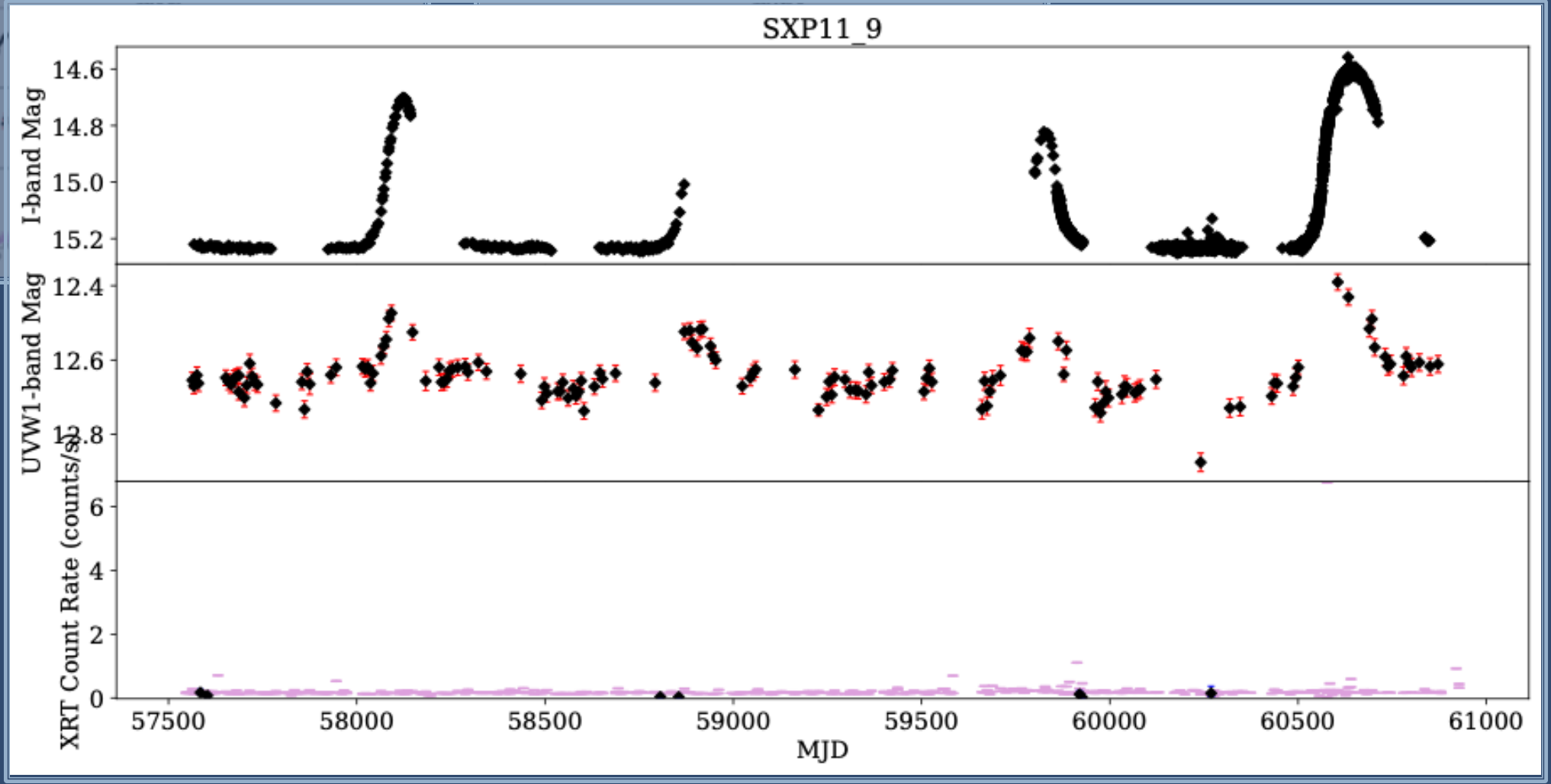

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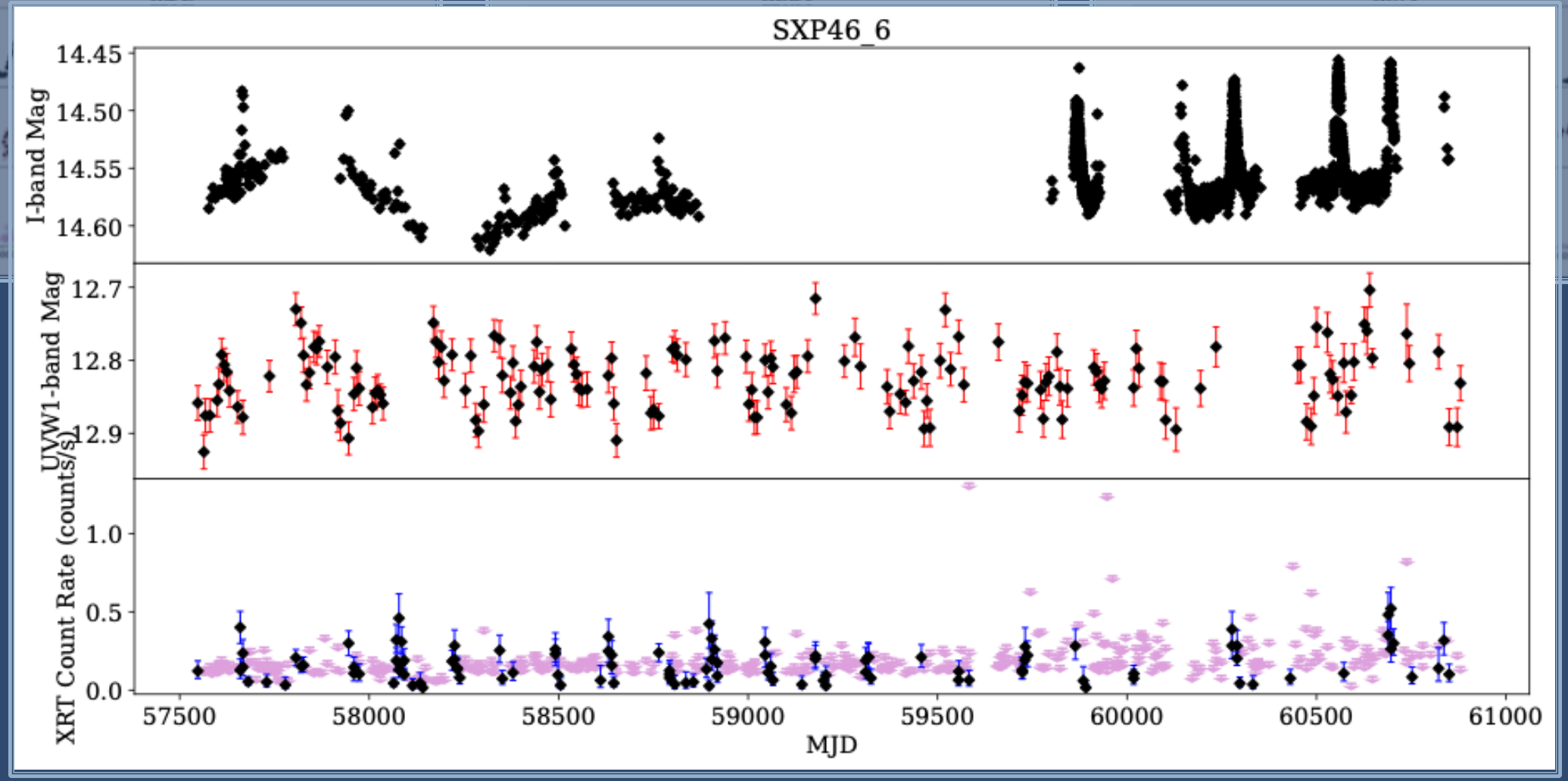

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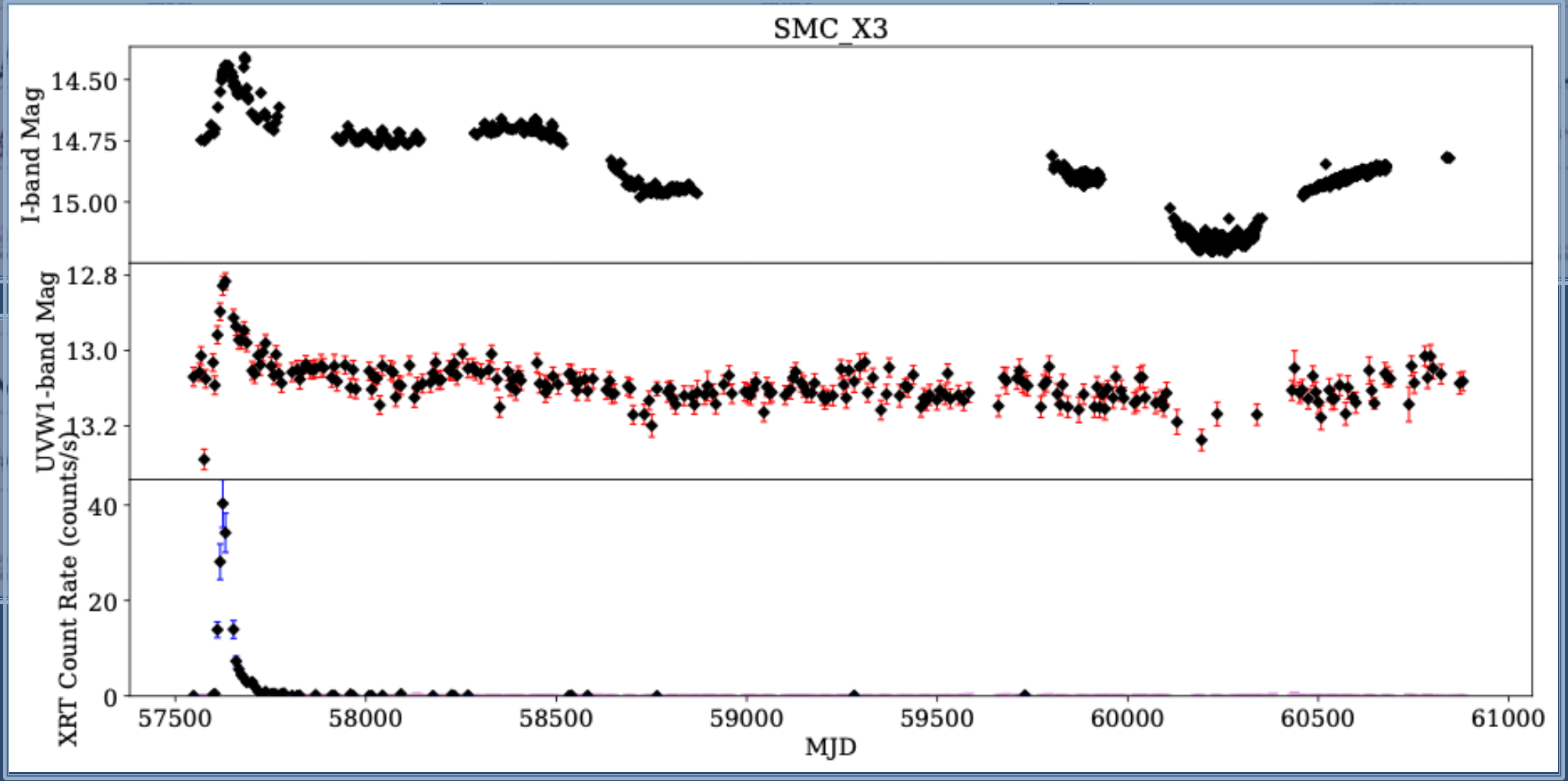

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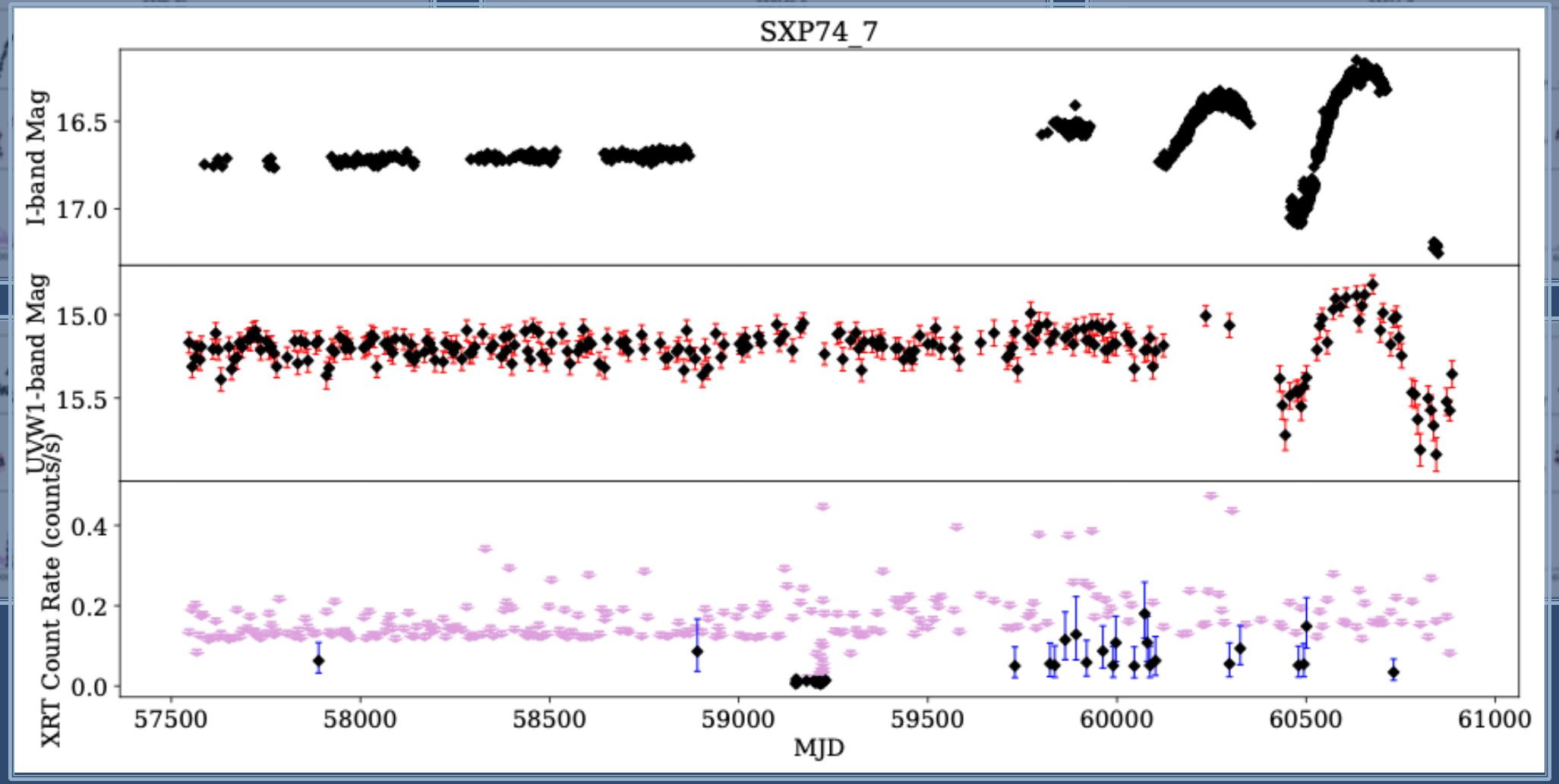

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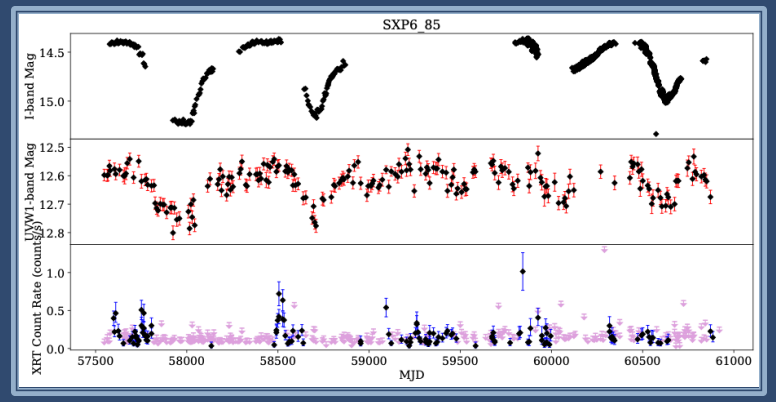

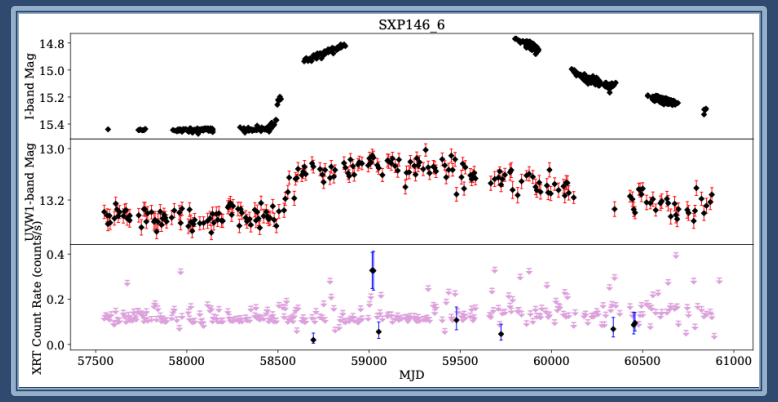

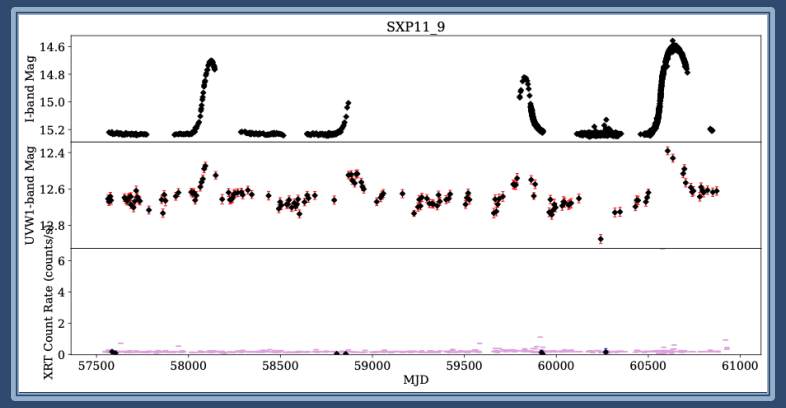

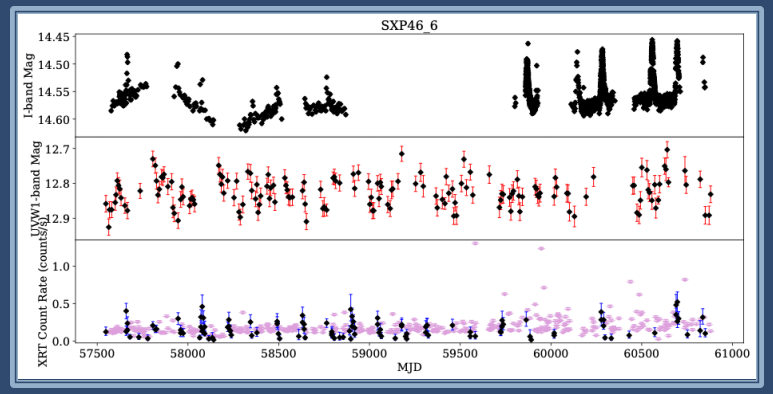

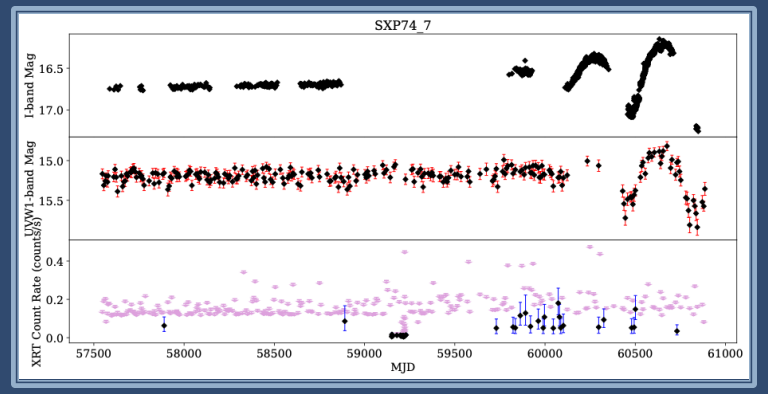

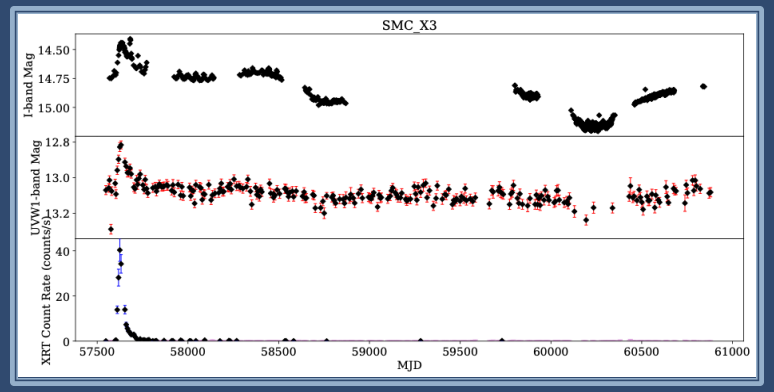

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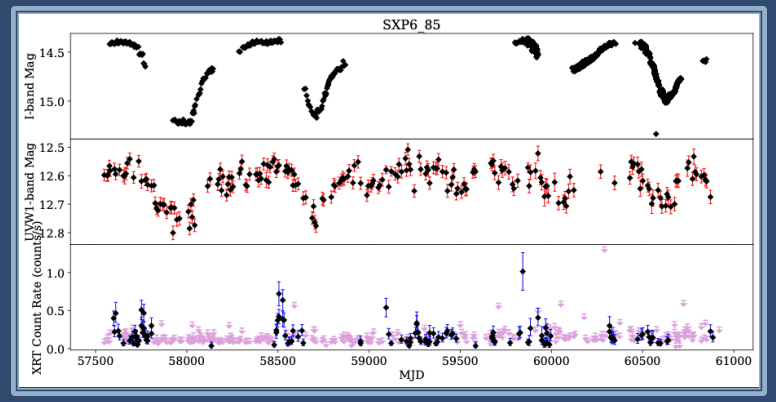
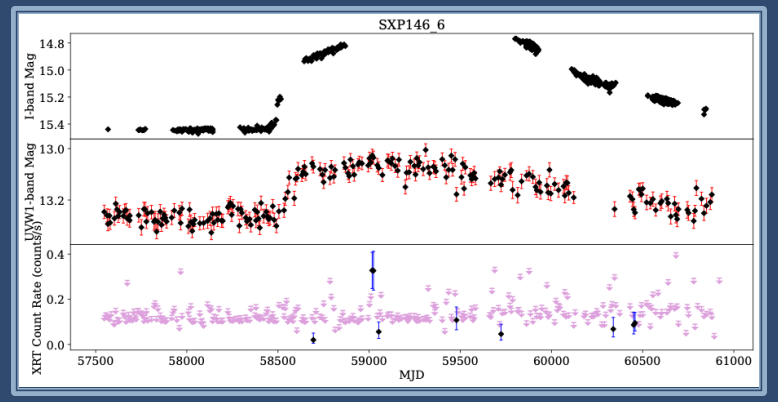
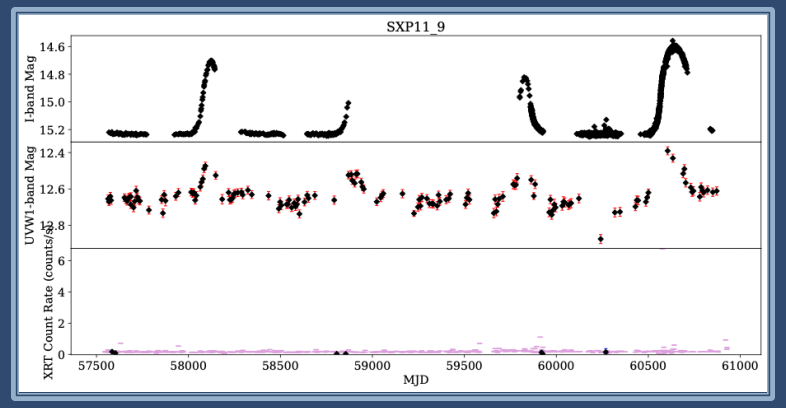
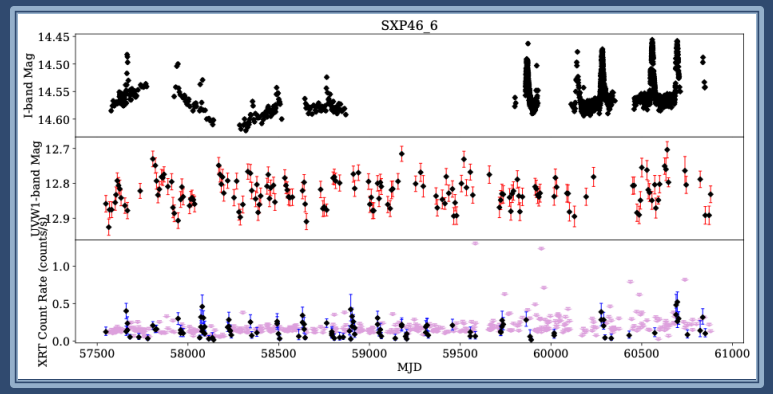
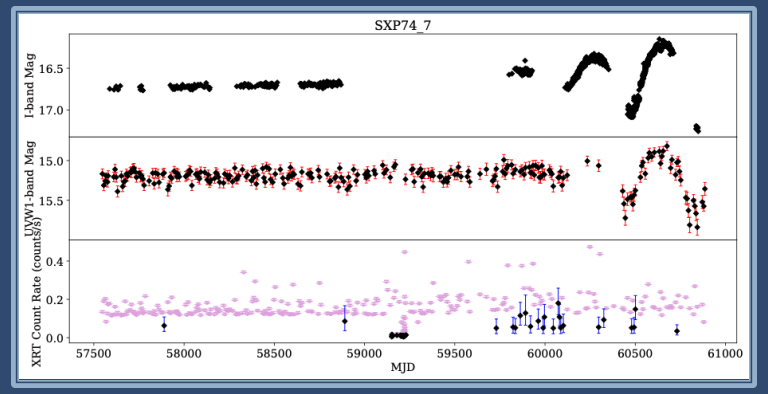
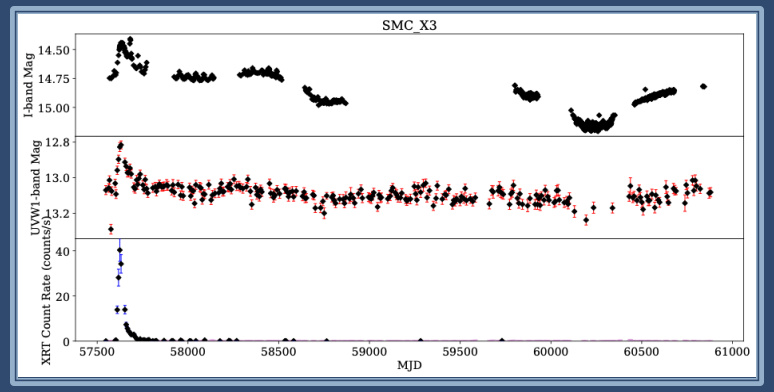












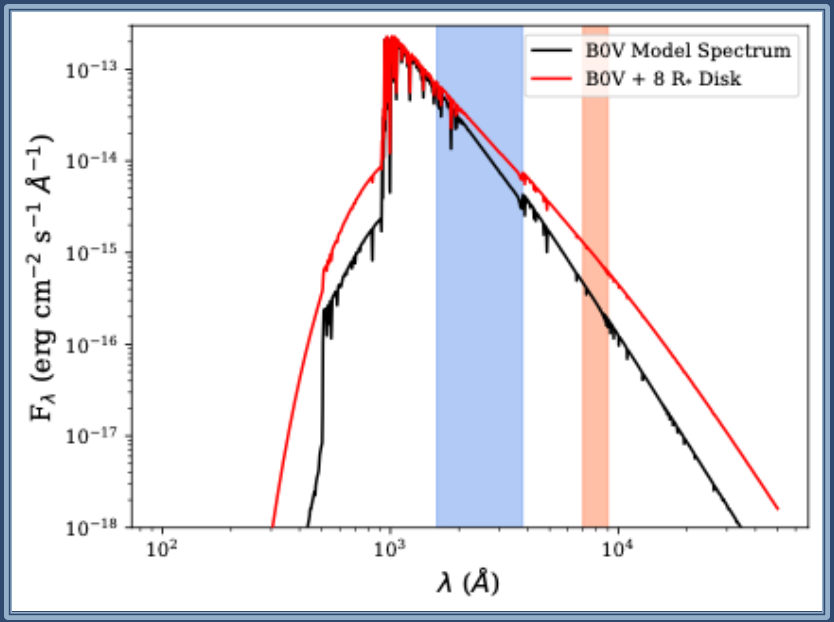
WHAT DRIVES UV VARIABILITY IN BE/X-RAY BINARIES?

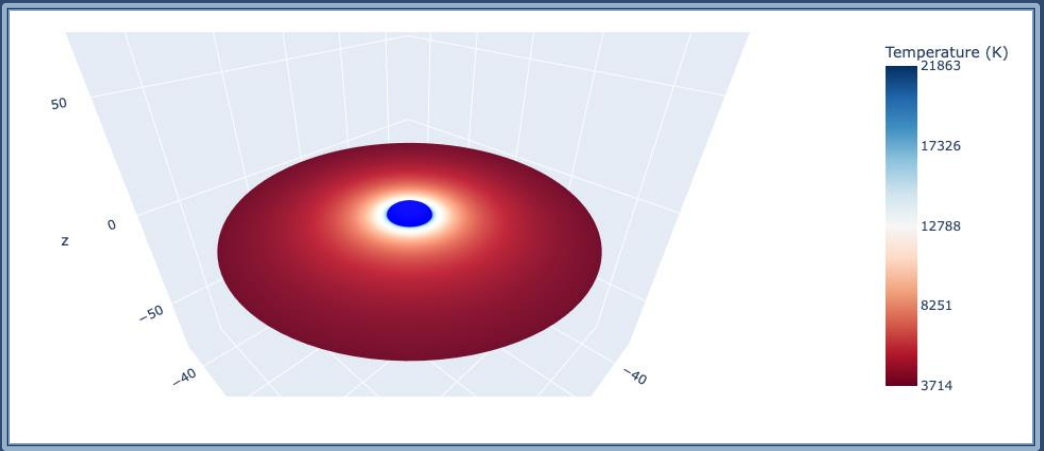
WHERE DOES IT ORIGINATE FROM IN THE SYSTEM?

Creating a Basic Model for UV Variability

- Start with Phoenix Models of O- and B-type stars at SMC metallicities ([M/H] = -1.0)
- Use the Viscous Decretion Disk Model to generate SEDs of Be stars (Carciofi & Bjorkman, 2006)
- Temperature Profile of the Disk:

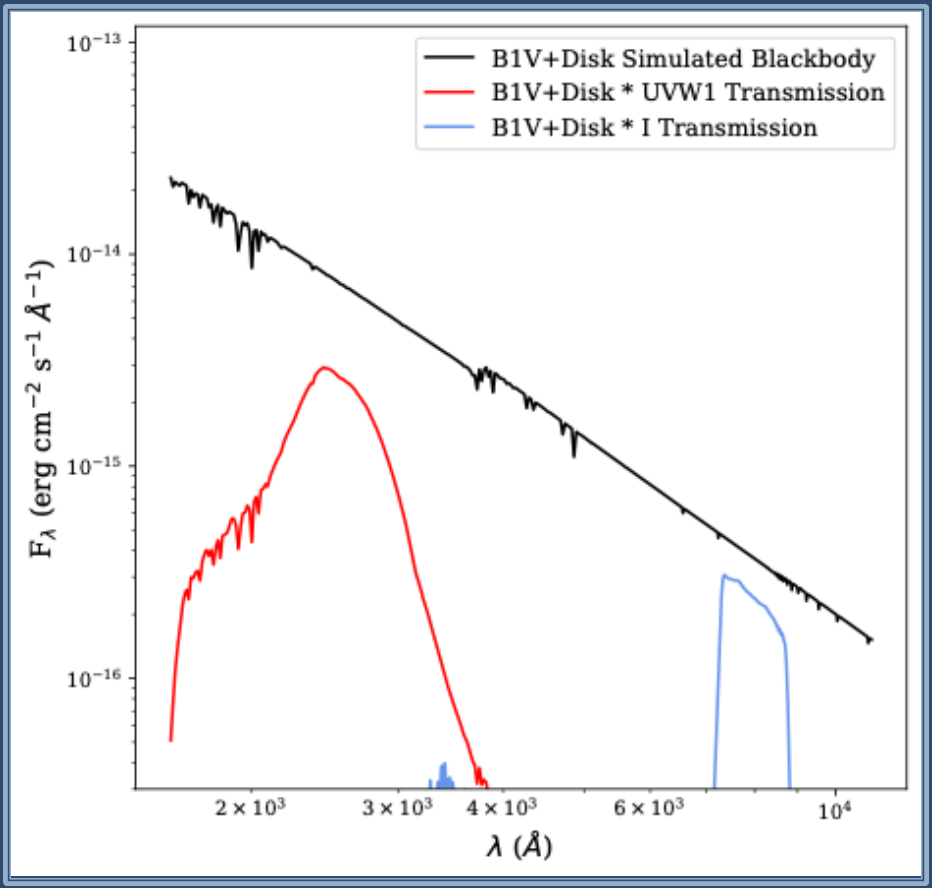
$$T(r) = \begin{cases} \dfrac{T_{Eff}}{\pi^{1/4}}\left(\sin^{-1}\left(\dfrac{R_*}{r}\right) - \left(\dfrac{R_*}{r}\right)\sqrt{1-\left(\dfrac{R_*}{r}\right)^2}\right)^{1/4}, & r \leq 10\,R_* \\ 0.9\,T_{Eff}\left(\left(\dfrac{r}{R_*}\right)^{3/4} - 10^{3/4}\right), & r > 10\,R_* \end{cases}$$



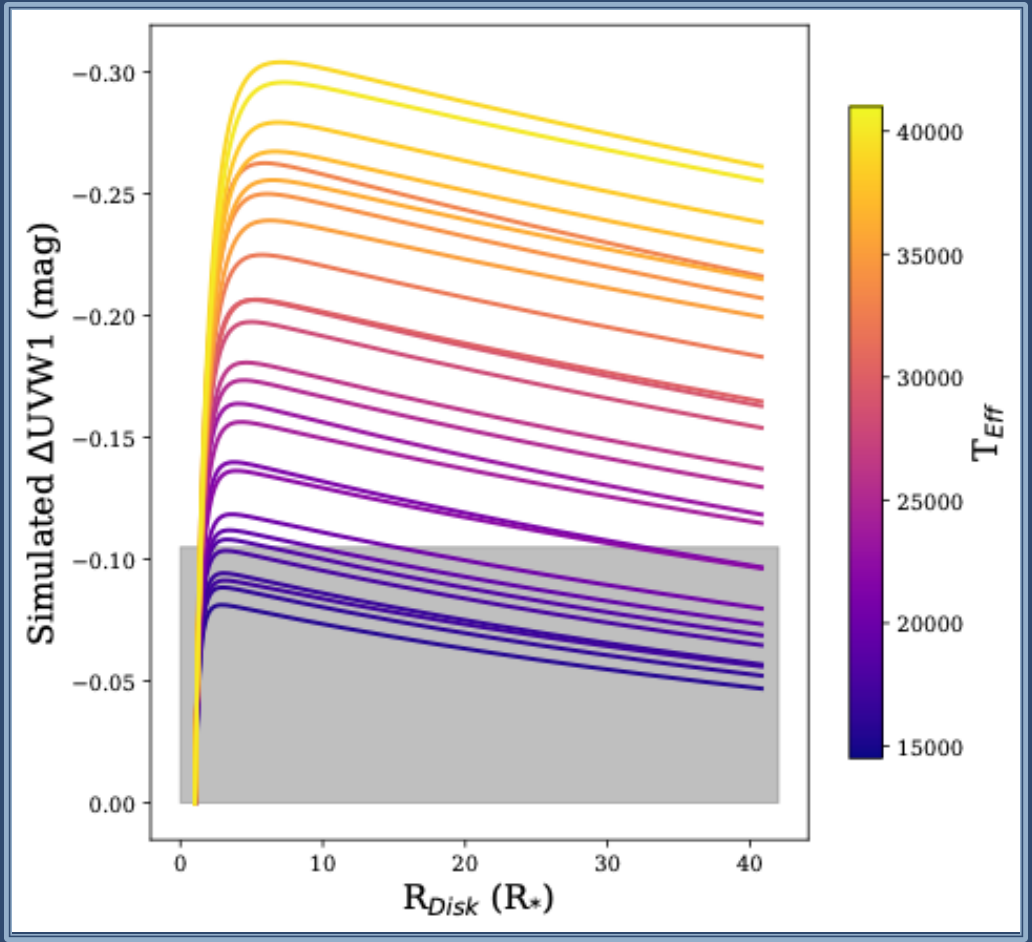


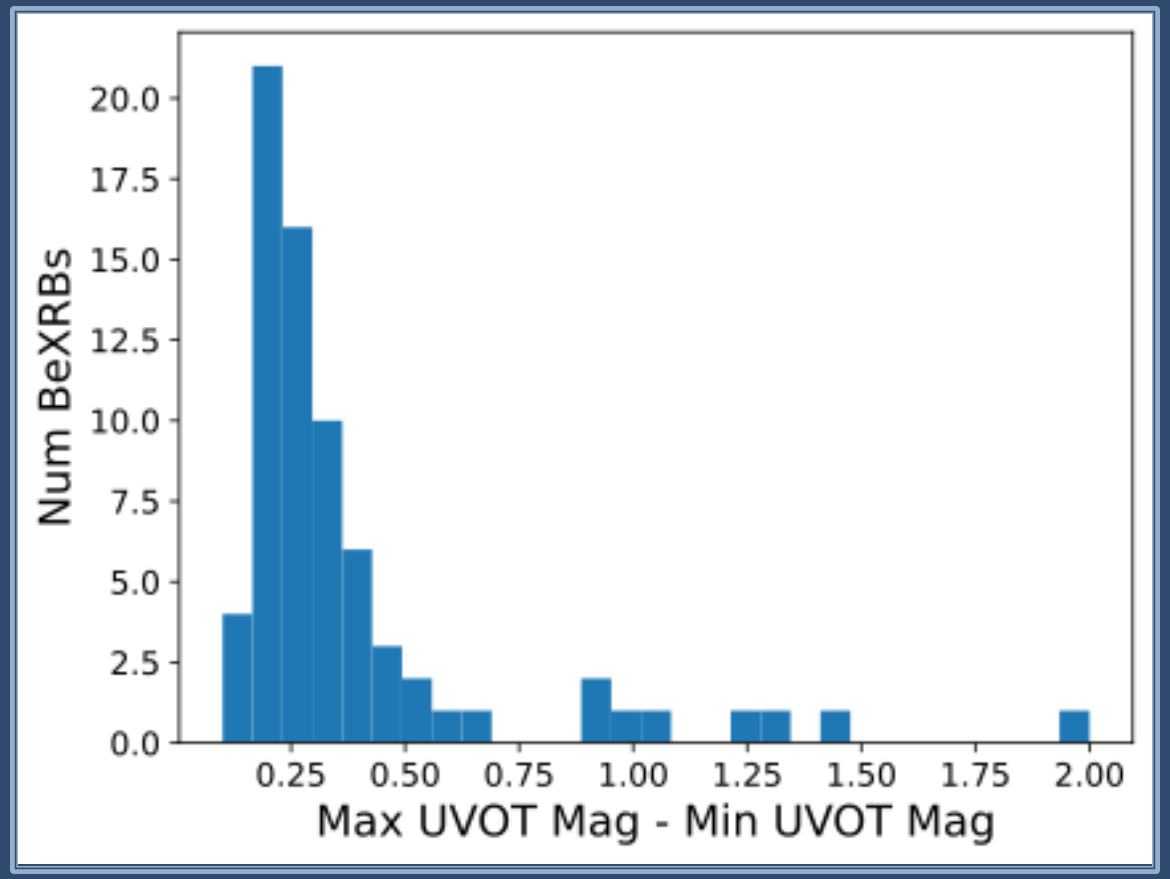
Creating a Basic Model for UV Variability

- Simplifying Assumptions:
  - Assume top-down view of the star ($i = 90°$)
  - Assume isothermal blackbody emission (no hot spots)
  - Assume geometrically thin disk
- Simulated Observations with Swift and OGLE:
  - Assume standard distance to SMC ($\mathrm{d} = 62.44$ kpc)
  - Assume average SMC extinction ($E_{V-I} \sim 1$)

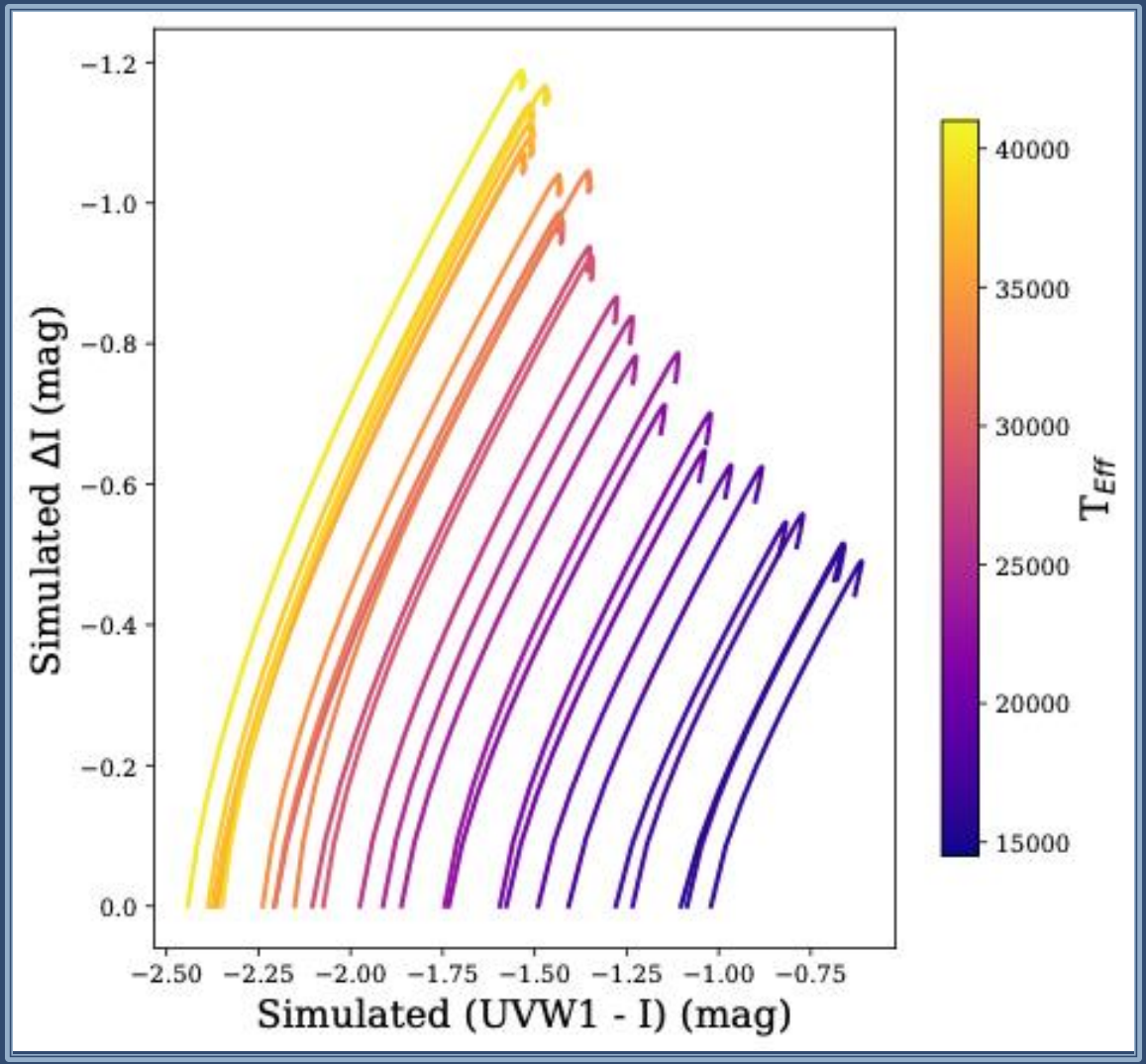


Results: UVW1 Magnitudes vs. Disk Radius

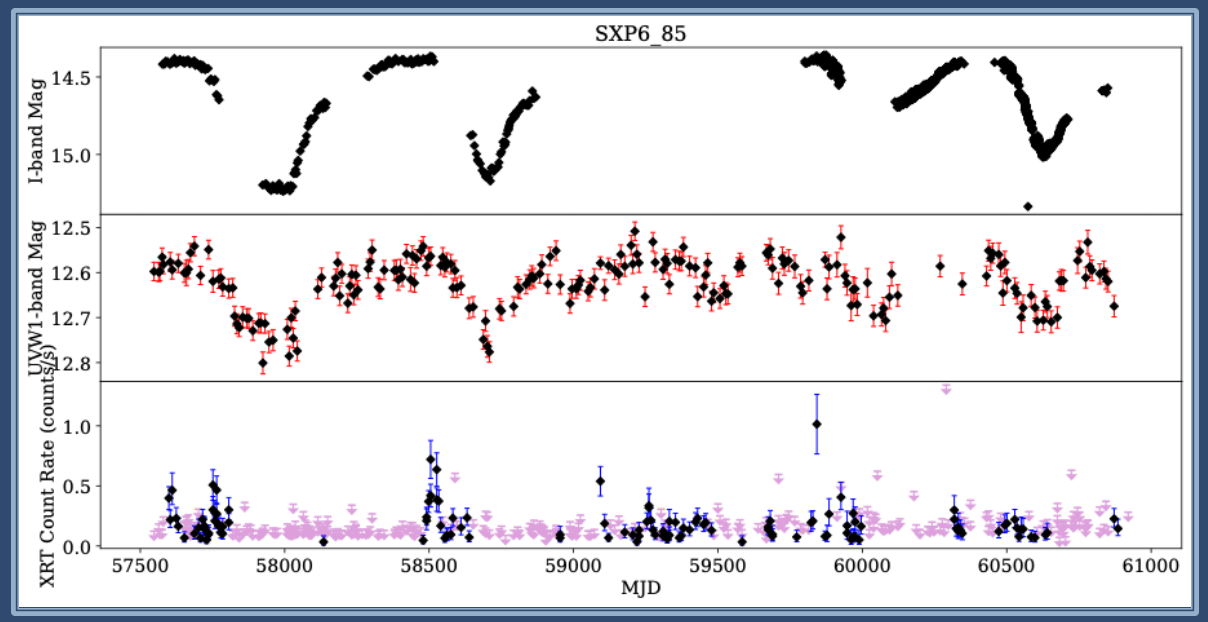


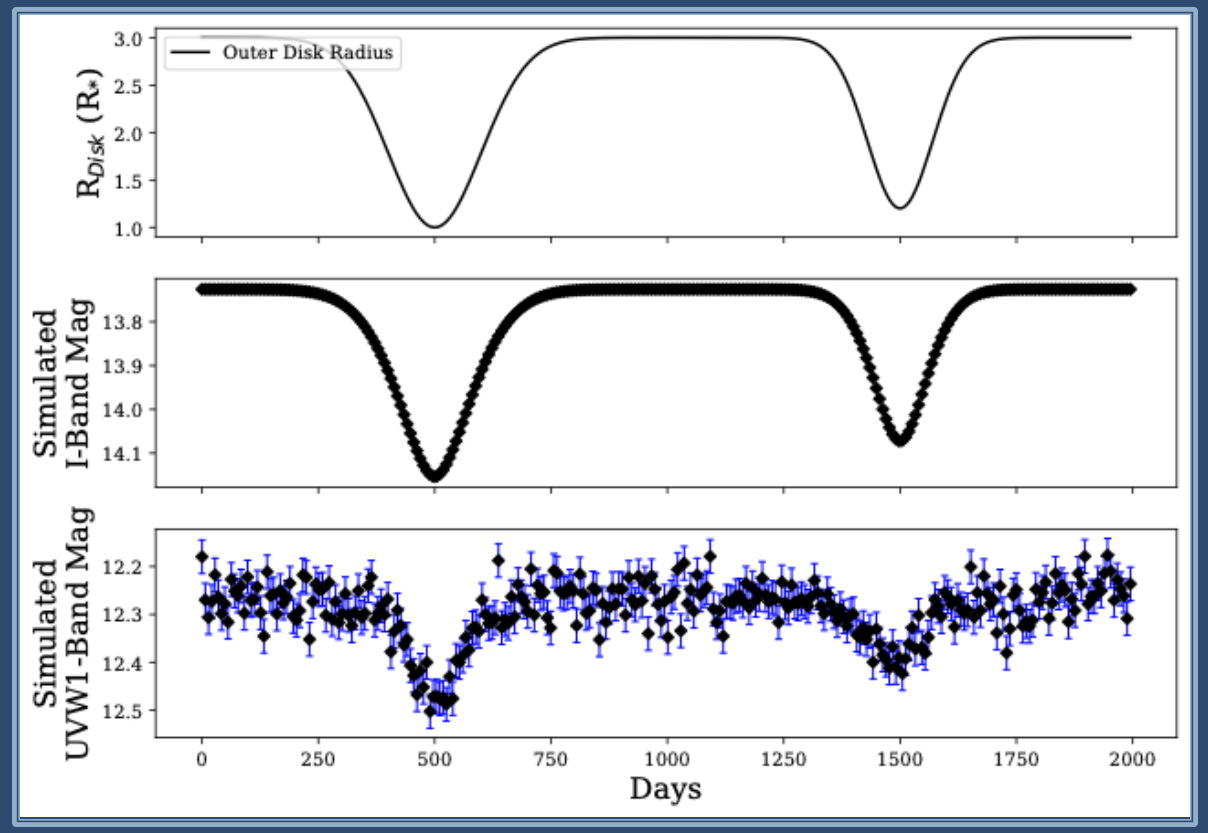


Results: Brighter When Redder

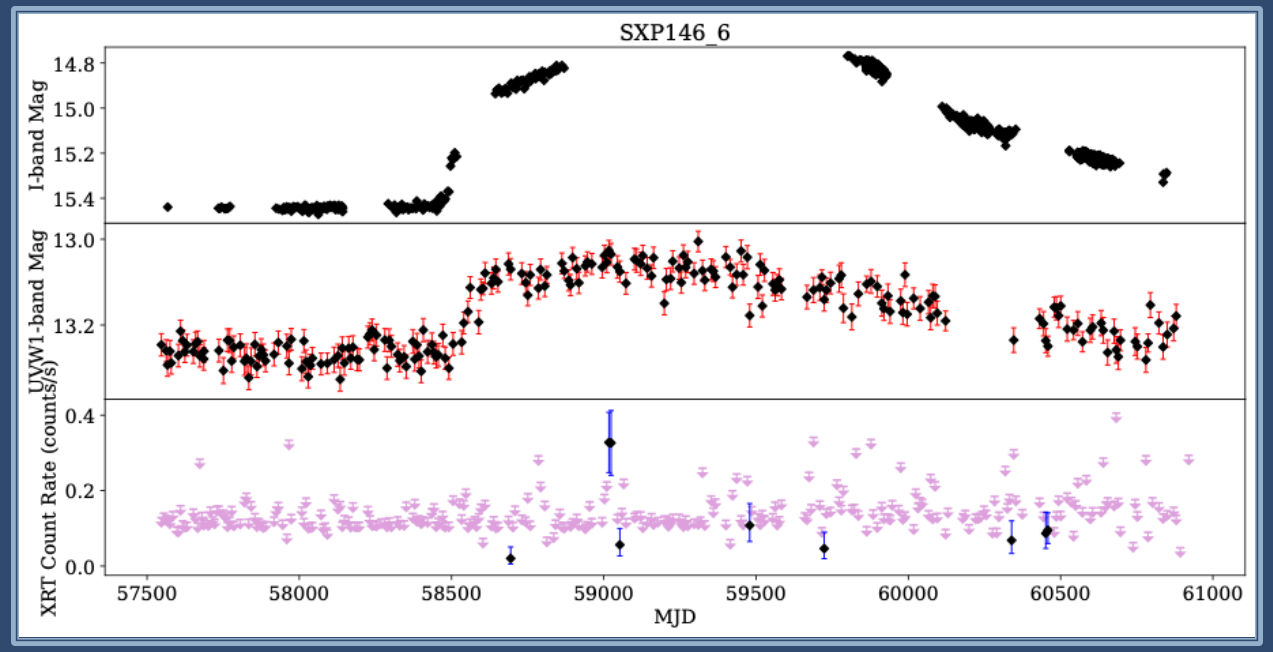


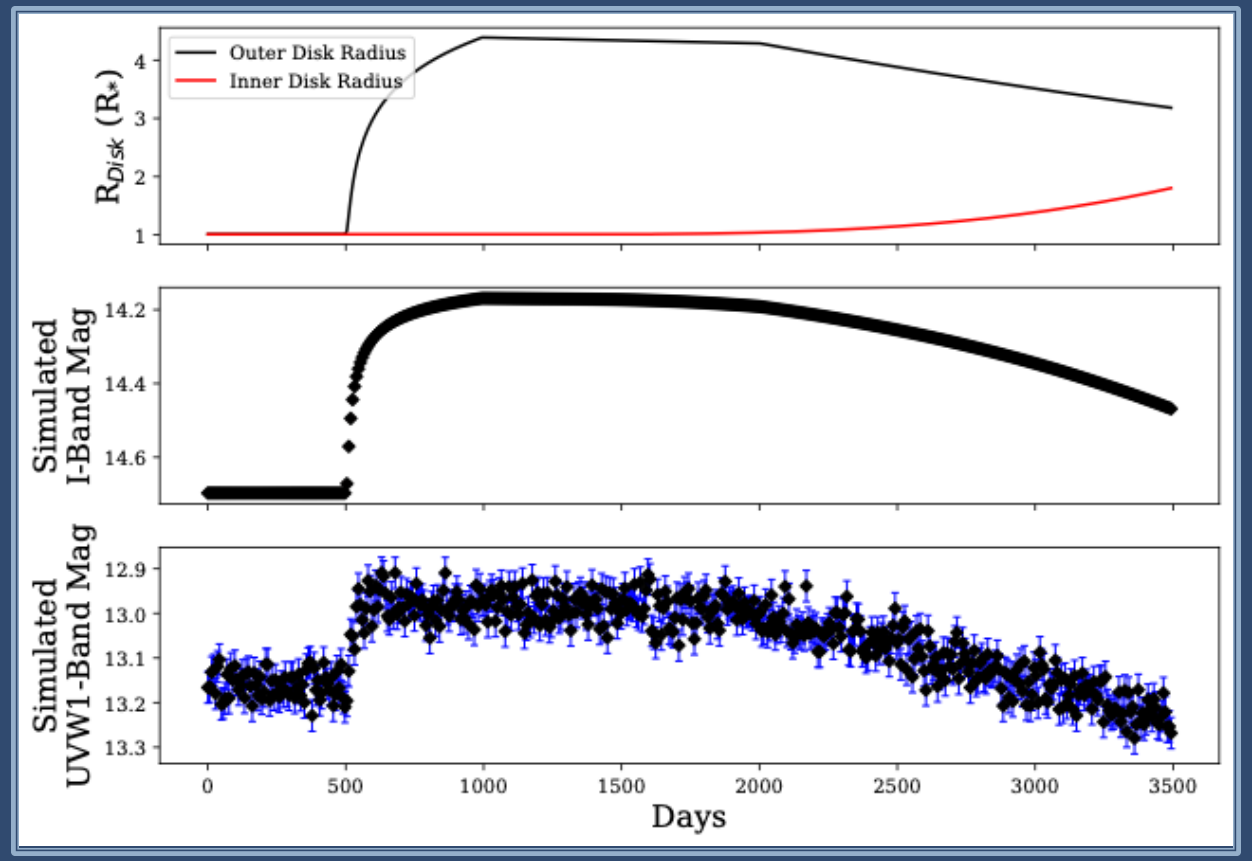
Application to Individual System: Disk Disruption by Companion





Application to Individual System: Disk Outburst Event





Conclusions and Future Work

Conclusions:

- S-CUBED data suggests that Be/X-ray binaries are variable at UV wavelengths.
- Variability can be recreated using the VDD Model in many cases.
- According to VDD:
  - UV bands probe changes in the innermost parts of a decretion disk.
  - 0.1 – 0.3 mag of variability expected from this region.

Future Work:

- Add in X-ray irradiation of the Be decretion disk by NS companion
- Allow disk to grow in scale height and add structure (e.g. spiral arms)
- Change inclination of the system
- Apply radiative transfer codes (HDUST)

Extra Slides

S-CUBED: The Swift SMC Survey

- Weekly Swift monitoring of the SMC
- 60s observations of 149 tiles
- Simultaneous UV and X-ray observations
  - **UVOT:** UVW1-band
  - **XRT:** 0.3-10 keV band

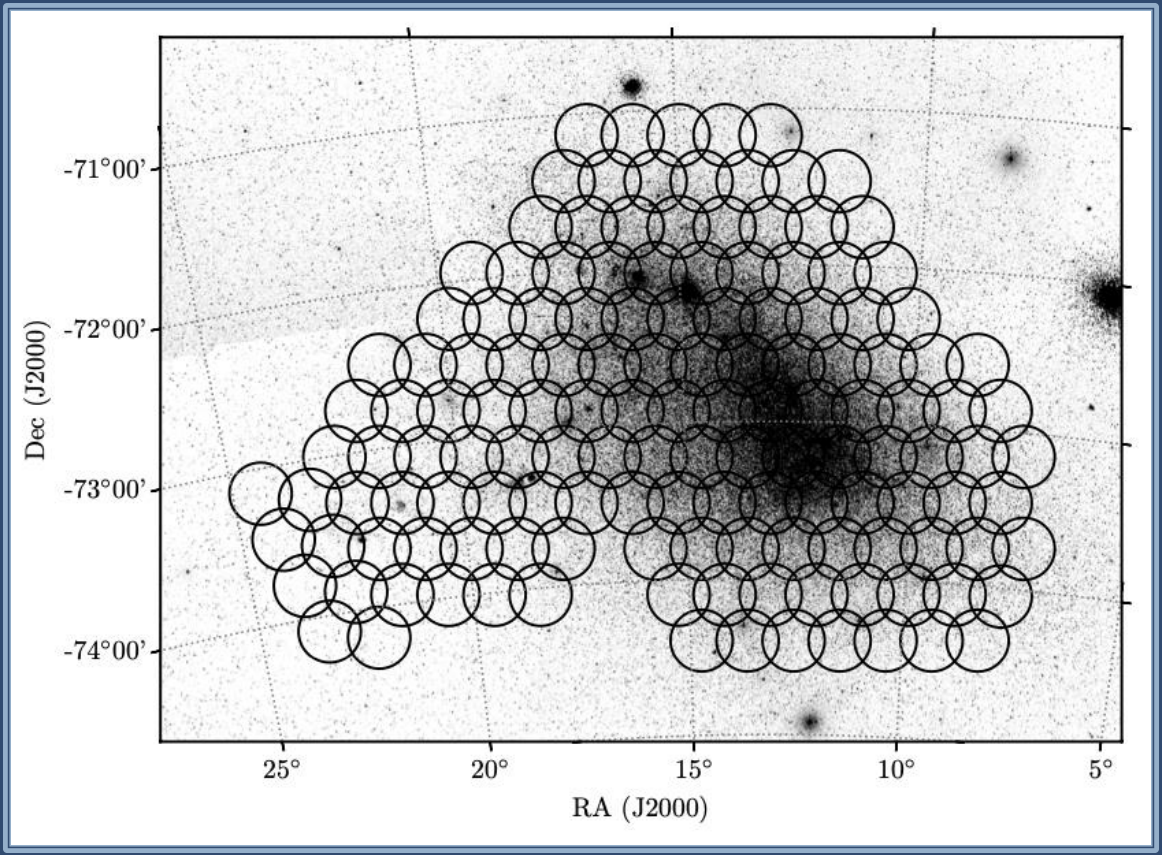


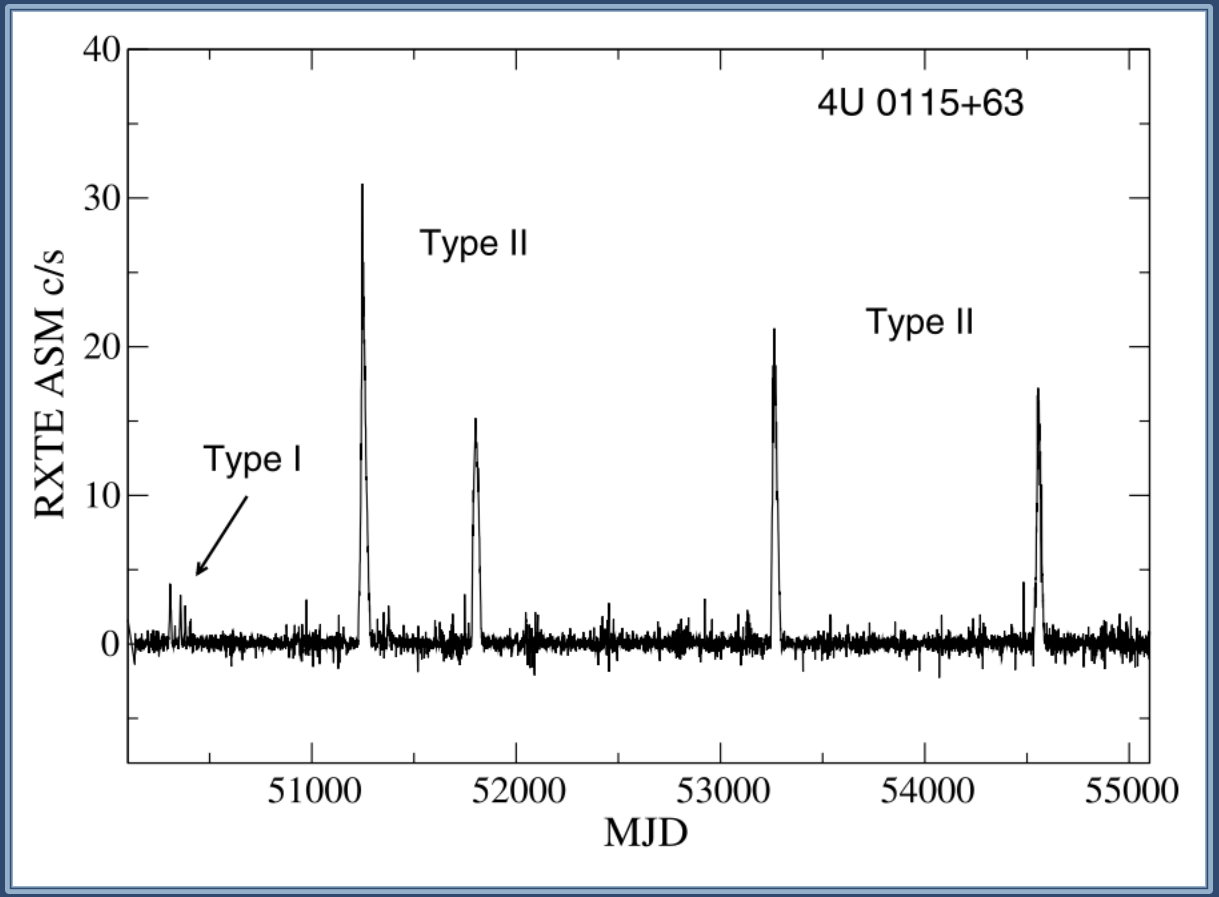
Image Credit: Kennea et al. 2018

Be/X-ray Binaries: The Standard Theory

- Transient X-ray emitters
- 2 Types of X-ray Outburst:
  - Type I

    $L_X \sim 10^{36} - 10^{37}$ erg s^{-1}
  - Type II

    $L_X \geq 10^{37}$ erg s^{-1}



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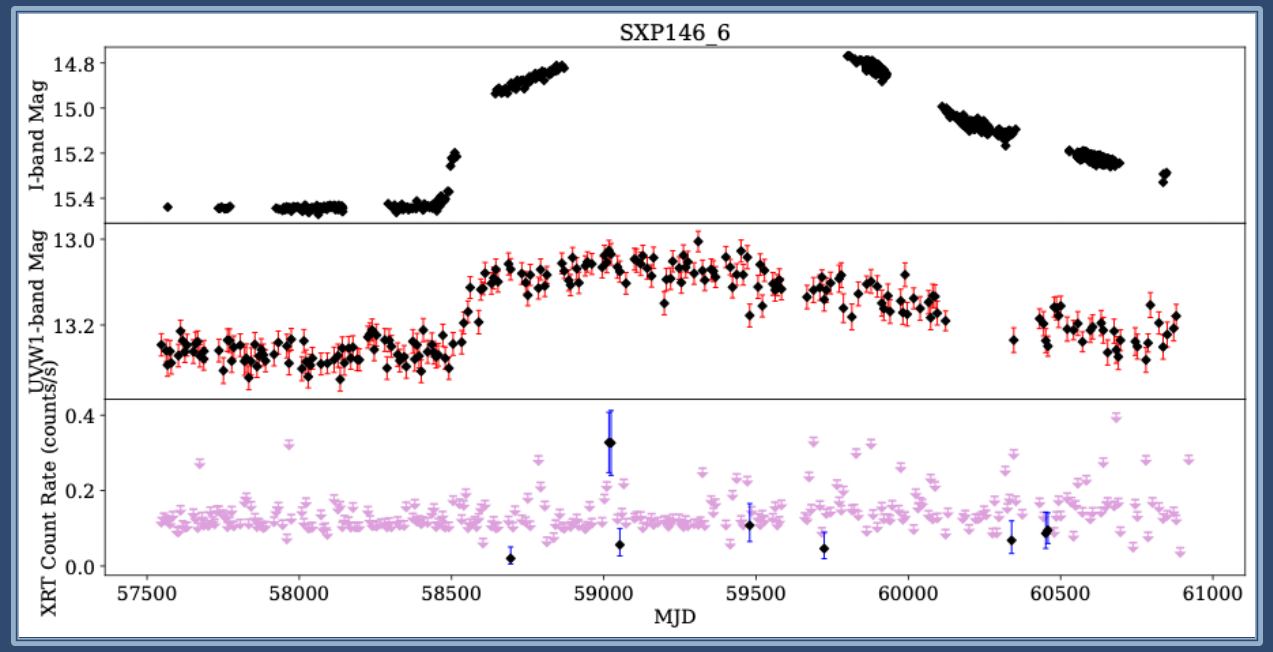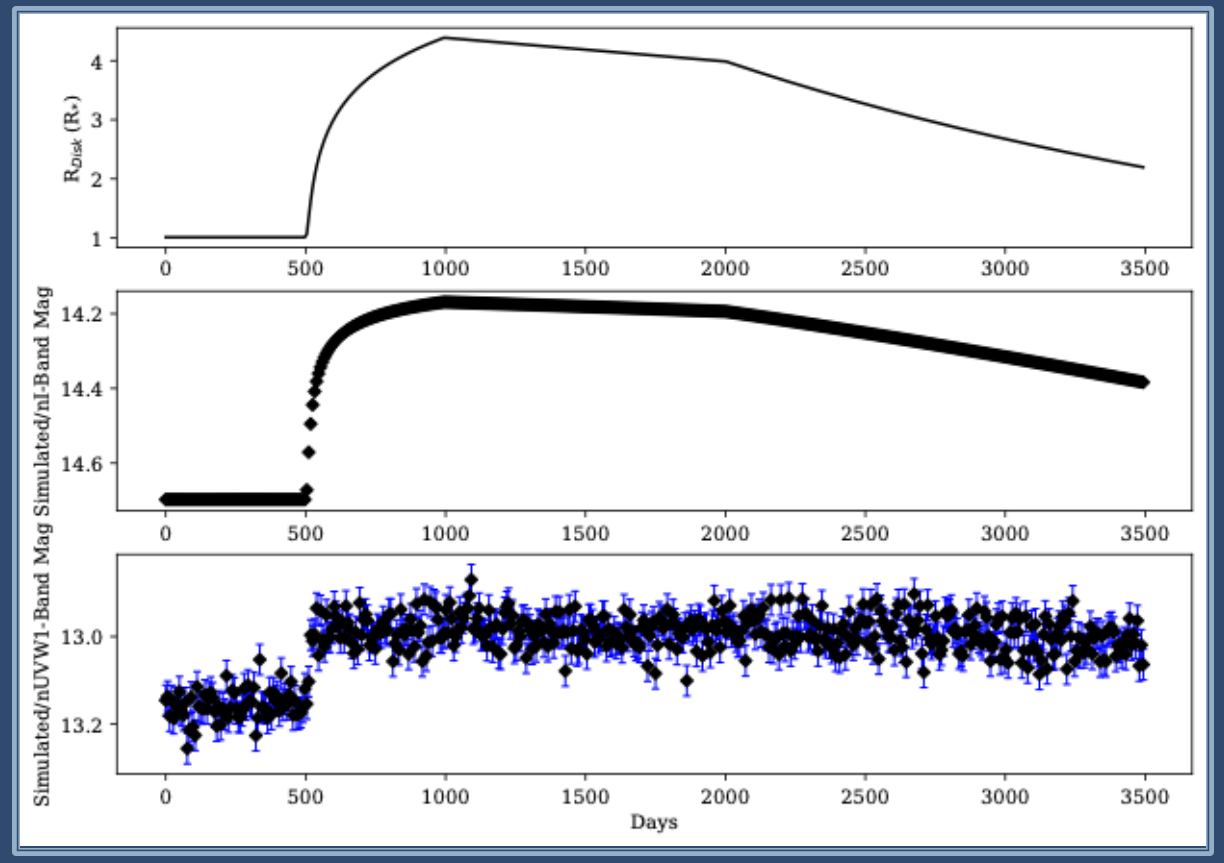
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